CSE 107 — Midterm 1 Practice

Fall 2026 55-minute attempt | 100 practice points

Name: _____ **Student ID:** _____

- For a timed attempt, use one page of notes (a cheat sheet) and a basic nonprogrammable calculator. Work independently.
- Answer all five questions in the spaces provided. Label any continuation with its question and part.
- For Questions 2–5, show your work. Numerical expressions may be left unevaluated.

1. (16 points)

Circle **T** or **F** for each statement. If false, give a brief reason, correction, or counterexample. True statements need no justification.

Each item is worth 2 points: a correct **T** earns 2; a correct **F** earns 1, plus 1 for a valid justification.

(a) If $P(A) = 2/5$ and $P(B) = 3/5$, then A and B are disjoint. **T** **F**

(b) If $P(A \cap B) = P(A)P(B)$, then A and B are independent. **T** **F**

(c) A random variable assigns exactly one real number to each outcome of an experiment. **T** **F**

(d) A proposed PMF on a finite set of values is valid whenever each listed probability lies between 0 and 1. **T** **F**

(e) If $F_X(2) = 3/4$, then $P(X > 2) = 1/4$. **T** **F**

(f) If a sample space has twelve outcomes and A contains three of them, then $P(A) = 1/4$. **T** **F**

(g) A group of three people can be chosen from eight people in $\binom{8}{3}$ ways. **T** **F**

(h) In a binomial model, the success probability may change from trial to trial. **T** **F**

2. (18 points)

The CDF of a discrete random variable X is

$$F_X(t) = \begin{cases} 0, & t < -1, \\ 1/10, & -1 \le t < 1, \\ 2/5, & 1 \le t < 3, \\ 4/5, & 3 \le t < 4, \\ 1, & t \ge 4. \end{cases}$$

(a) List the possible values of X and give its PMF $p_X(x)$ for all real x. **[8 points]**

(b) Find $P(X < 1)$ and $P(-1 < X \le 3)$. **[10 points]**

3. (20 points)

A code consists of three letters chosen from A, B, C, D, E. Letters may repeat, and every three-letter code is equally likely.

(a) How many codes contain exactly two different letters? **[10 points]**

(b) Given that the first and last letters are different, what is the probability that the code contains exactly two different letters? **[10 points]**

4. (22 points)

For a randomly selected package, let E be the event that it was shipped express and let D be the event that it was delivered the next day. Suppose

$$P(E) = \frac{2}{5}, \qquad P(D \mid E) = \frac{3}{5}, \qquad P(E \mid D) = \frac{4}{5}.$$

(a) Find $P(D)$. **[6 points]**

(b) Find $P(D \mid E^c)$. **[6 points]**

(c) Find the probability that exactly one of the events E and D occurs. **[6 points]**

(d) Are E and D independent? Justify your answer. **[4 points]**

5. (24 points)

A sensor makes six independent readings. Each reading is incorrect with probability 1/5. Let N be the number of incorrect readings.

(a) State the distribution of N, including its parameter values and possible values, and find $P(N \geq 5)$. **[8 points]**

(b) Given that exactly two readings are incorrect, what is the probability that the first two readings are both correct? **[8 points]**

(c) For this part only, suppose the sixth reading is incorrect with probability 1/2. The other five readings still have error probability 1/5, and all six readings are independent. Find the probability that all six readings are correct. Is $\text{Bin}(6, 1/5)$ a valid model for N in this case? Briefly explain. **[8 points]**